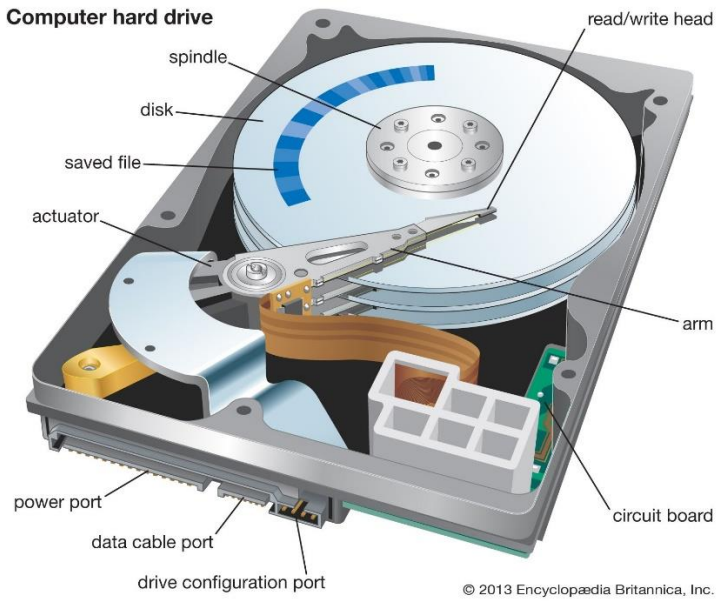
Lecture 9 - Secondary Storage & Modern Architectures

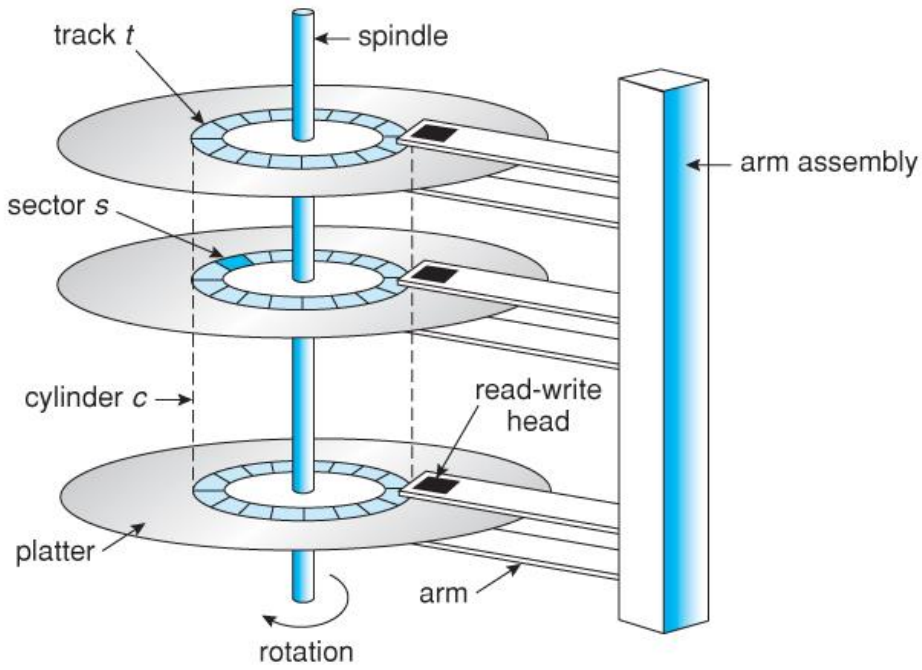
1. Secondary Storage

Secondary storage, also known as auxiliary storage or external memory, is non-volatile memory that retains data even when power is off. It is used for long-term storage of data and programs that are not currently being used by the CPU. Unlike main memory (RAM), secondary storage is typically slower but offers significantly higher capacity and lower cost per bit.

1.1 Magnetic Disks (Hard Disk Drives - HDDs)

- **Concept:** HDDs store data magnetically on rapidly spinning platters coated with a magnetic material.
- **Architecture:**
  - **Platters:** Circular disks made of glass or aluminum, coated with magnetic material on both sides. Multiple platters are stacked on a central spindle.
  - **Spindle:** Rotates the platters at high speeds (e.g., 5,400 to 15,000 RPM).
  - **Read/Write Heads:** Tiny electromagnets, one for each platter surface, mounted on an arm. They float on a cushion of air just nanometers above the platter surface.
  - **Actuator Arm:** Moves the read/write heads across the platter surfaces to access different tracks.
  - **Tracks:** Concentric circles on the platter surface where data is stored.
  - **Sectors:** Divisions of a track, typically 512 bytes or 4KB, representing the smallest unit of data that can be read or written.
  - **Cylinders:** A set of all tracks that are at the same radial distance on all platters.





- **How it Works:**
  - **Writing:** The read/write head magnetizes tiny regions on the platter surface, with the direction of magnetization representing binary 0s and 1s.
  - **Reading:** The read/write head detects the magnetic fields generated by these magnetized regions, inducing small electrical currents that are then interpreted as data.
- **Access Time Components:**
  - **Seek Time:** The time it takes for the read/write heads to move to the correct track. This is the most significant component of HDD access time.
  - **Rotational Latency:** The time it takes for the desired sector to rotate under the read/write head.
  - **Transfer Time:** The time it takes to actually transfer the data once the head is positioned correctly.
- **Characteristics:**
  - **Non-volatile:** Data persists without power.
  - **High Capacity:** Can store terabytes of data (e.g., 2TB, 4TB, 8TB, 16TB).
  - **Low Cost per GB:** Historically the cheapest storage per unit of capacity.
  - **Mechanical Parts:** Susceptible to physical shock and wear, generate noise and heat.

- **Relatively Slow:** Due to mechanical movement, access times are in milliseconds.

1.2 Solid State Drives (SSDs) - Architecture and Advantages

- **Concept:** SSDs store data using non-volatile flash memory (typically NAND flash memory). They have no moving parts.
- **Architecture:**
  - **Flash Memory Cells:** Data is stored in electrical charges within floating-gate transistors.
  - **Controller:** A sophisticated processor that manages data storage, retrieval, error correction, wear leveling, and garbage collection. This is crucial for SSD performance and longevity.
  - **Cache (DRAM):** Many SSDs include a small amount of DRAM to act as a cache for mapping tables, speeding up access.
  - **Interface:** Connects to the system via interfaces like SATA (SATA III, up to ~600 MB/s), PCIe (NVMe protocol, significantly faster, e.g., thousands of MB/s).
- **How it Works:** Data is written to and read from blocks of flash memory cells. The controller manages the complex process of organizing, erasing, and writing data to ensure performance and longevity.
- **Advantages over HDDs:**
  - **Speed:** Significantly faster read/write speeds (sequential and random) due to no mechanical parts. This translates to faster boot times, application loading, and file transfers.
  - **Durability:** Much more robust and resistant to physical shock as there are no moving parts. Ideal for laptops and portable devices.
  - **No Noise:** Silent operation.
  - **Lower Power Consumption:** Leads to better battery life in laptops and less heat generation.
  - **Smaller Form Factors:** Available in very compact sizes (e.g., M.2).
  - **Lower Latency:** Data access is almost instantaneous compared to HDDs.
- **Disadvantages:**
  - **Higher Cost per GB:** Historically more expensive than HDDs, though prices are decreasing.

- **Limited Write Endurance:** Flash memory cells have a finite number of program/erase cycles. The controller uses techniques like wear leveling to distribute writes evenly and prolong lifespan, but it's still a factor for extremely heavy write workloads. Modern SSDs have very long lifespans for typical consumer use.

1.3 Optical Storage (CD, DVD, Blu-ray - brief)

- **Concept:** Optical storage devices use lasers to read and write data by detecting variations in light reflection from a disc surface.
- **How it Works (General Principle):**
  - Data is encoded as microscopic "pits" (depressions) and "lands" (flat areas) on the disc surface.
  - A laser beam is shone onto the disc.
  - When the laser hits a land, it reflects back directly. When it hits a pit, the light is diffracted and does not reflect back as strongly.
  - A photosensor detects these differences in reflection, which are then translated into binary 0s and 1s.
  - For writable discs (CD-R, DVD-R, BD-R), a high-power laser "burns" pits into a dye layer. For rewritable discs (CD-RW, DVD-RW, BD-RE), the laser changes the phase of a special material.

- **Types :**
    - **CD (Compact Disc):** Uses a red laser. Typical capacity ~700 MB. Primarily used for audio and early software distribution.
    - **DVD (Digital Versatile Disc):** Uses a red laser, but with a shorter wavelength and more advanced encoding, allowing for higher density. Typical capacity ~4.7 GB (single-layer) to 8.5 GB (dual-layer). Popular for video (movies) and software.
    - **Blu-ray Disc:** Uses a blue-violet laser, which has an even shorter wavelength, enabling even higher data density. Typical capacity ~25 GB (single-layer) to 50 GB (dual-layer) or more. Designed for high-definition video and large data archiving.
- **Characteristics:**
    - **Non-volatile:** Data persists without power.
    - **Removable:** Easily portable.
    - **Archival:** Good for long-term storage if stored properly.
    - **Relatively Slow Access:** Slower than HDDs for random access, but good for sequential reads.
    - **Lower Capacity (compared to HDDs/SSDs):** Limited by physical size and laser wavelength.

1.4 Introduction to Cloud Storage (Conceptual)

- **Concept:** Cloud storage is a model where digital data is stored in logically pooled physical storage, typically across multiple servers, and the physical environment is owned and managed by a third-party provider (e.g., Google Cloud, AWS, Microsoft Azure). Users access their data over the internet.
- **How it Works (High-Level):**
    1. Users upload data to a cloud storage provider via an internet connection.
    2. The provider stores this data on their vast infrastructure of servers, often replicating it across multiple data centers for redundancy and availability.
    3. Users can access their data from anywhere, on any device, with an internet connection, using web portals, APIs, or specialized applications.
- **Key Characteristics:**
    - **Scalability:** Users can easily increase or decrease their storage capacity on demand.

- **Accessibility:** Data can be accessed from virtually anywhere with an internet connection.
- **Durability and Redundancy:** Cloud providers typically replicate data across multiple locations to ensure high availability and prevent data loss.
- **Cost-Effectiveness:** Often a pay-as-you-go model, eliminating the need for users to purchase and maintain their own physical storage infrastructure.
- **Managed Service:** The provider handles maintenance, security, and updates of the storage infrastructure.

- **Types of Cloud Storage (brief mention):** Object storage (for unstructured data), File storage (for network shared folders), Block storage (for virtual disks).

2. Modern Architectures (Brief Introduction)

Beyond traditional CPU-centric designs, modern computing systems increasingly incorporate specialized processors to handle diverse computational tasks more efficiently.

2.1 Multi-core Processors

- **Concept:** Instead of having a single processing unit (core) on a chip, a multi-core processor integrates two or more independent processing units (cores) onto a single integrated circuit (die).
- **How it Works:** Each core is a complete processing unit with its own Arithmetic Logic Unit (ALU), control unit, and registers, and often its own L1/L2 cache. All cores typically share a larger L3 cache and the same main memory interface.
- **Advantages:**
  - **Parallelism:** Allows a single CPU to execute multiple instruction threads concurrently, significantly improving performance for parallelizable workloads (e.g., running multiple applications simultaneously, video encoding, scientific simulations).
  - **Improved Multitasking:** The operating system can schedule different processes or threads to run on different cores, leading to a smoother and more responsive user experience.
  - **Power Efficiency:** Often more power-efficient than achieving similar performance with a single, faster core.

- **Challenges:**
    - **Software Parallelization:** Not all software is designed to take full advantage of multiple cores. Programs need to be written to be "multithreaded" to benefit from parallel execution.
    - **Cache Coherence:** Maintaining consistency across multiple cores' caches is a complex hardware/software problem.

2.2 GPUs (Graphics Processing Units) for General-Purpose Computing

- **Concept:** GPUs were originally designed to accelerate graphics rendering by performing highly parallel computations on pixels. Modern GPUs are now also used for General-Purpose computing on Graphics Processing Units (GPGPU), leveraging their parallel architecture for a wide range of non-graphics tasks.
- **Architecture:**
    - **Massively Parallel:** Unlike CPUs with a few powerful, complex cores, GPUs have hundreds or even thousands of simpler, smaller processing cores (often called Stream Processors or CUDA Cores).
    - **Specialized for Parallel Data:** Optimized for performing the same operation on many different data elements simultaneously (e.g., matrix multiplications, vector operations).
- **How it's Used in GPGPU:**
    - Programmers use specialized programming models and APIs (like NVIDIA's CUDA or OpenCL) to offload highly parallelizable computations from the CPU to the GPU.
    - The CPU prepares data and commands, sends them to the GPU, and then the GPU executes the parallel computations much faster than a CPU could.
- **Advantages for GPGPU:**
    - **Immense Speedup:** Can provide orders of magnitude speedup for workloads that are highly parallel (e.g., scientific simulations, machine learning training, cryptocurrency mining, video processing).
    - **Energy Efficiency for Parallel Tasks:** More energy-efficient for specific parallel tasks than CPUs.
- **Use Cases:** Artificial Intelligence (AI) and Machine Learning (ML), scientific computing, data analytics, cryptography, financial modeling, high-performance computing (HPC).

2.3 Brief mention of Emerging Architectures (Very High Level)

The quest for faster, more efficient, and specialized computing continues, leading to research in novel architectures:

- **Neuromorphic Computing:**
  - **Concept:** Architectures inspired by the structure and function of the human brain.
  - **Goal:** To achieve highly energy-efficient and fault-tolerant computing for AI workloads, especially for tasks like pattern recognition and learning.
  - **Approach:** Focuses on massive parallelism, event-driven processing, and colocation of memory and processing elements.
  - **Examples:** IBM's TrueNorth, Intel's Loihi.
- **Quantum Computing:**
  - **Concept:** A fundamentally different paradigm of computing that leverages principles of quantum mechanics (superposition, entanglement) to perform calculations.
  - **Basic Unit:** Qubit (can represent 0, 1, or a superposition of both simultaneously).
  - **Potential:** Could solve certain complex problems (e.g., drug discovery, materials science, cryptography, optimization) that are intractable for even the most powerful classical supercomputers.
  - **Current Status:** Still in early research and development, facing significant engineering challenges (maintaining qubit coherence, error correction). Not a replacement for classical computers for general-purpose tasks, but a specialized tool for specific problems.

Activities: Discussion on Pros and Cons of Different Storage Technologies

Let's discuss the advantages and disadvantages of the main secondary storage technologies we've covered:

Feature	Magnetic Disks (HDDs)	Solid State Drives (SSDs)	Optical Storage (CD/DVD/Blu-ray)	Cloud Storage
Pros	- High capacity - Low cost per GB - Good for bulk, infrequent storage	- Very fast - Durable (no moving parts) - Silent - Low power consumption - Small form factors	- Removable/portable - Good for distribution - Archival (if stored well) - Immune to magnetic fields	- Highly scalable - Accessible anywhere - Managed by provider - Redundant/durable - Pay-as-you-go
Cons	- Slow (mechanical) - Fragile (moving parts) - Noisy - Higher power consumption	- Higher cost per GB - Limited write endurance (less of an issue now) - Data recovery can be harder if controller fails	- Lower capacity - Slower random access - Susceptible to scratches/damage - Requires dedicated drives	- Requires internet connection - Security/privacy concerns (trusting third party) - Latency dependent on network - Vendor lock-in potential - Ongoing costs

Discussion Questions:

1. **Scenario 1:** You are building a high-performance gaming PC. Which storage technology would you choose for the operating system and games, and why? Which for large media files (movies, photos) and why?
2. **Scenario 2:** You need to back up critical business data for long-term archival, but access is rare. Which technology would be most appropriate, considering cost and longevity?
3. **Scenario 3:** A new startup wants to deploy an application that requires flexible and rapidly scalable storage for user-generated content. What is the most suitable option?
4. How do the advantages of multi-core processors and GPUs complement each other in a modern computing system? Give examples of tasks best suited for each.
5. Why are quantum and neuromorphic computing considered "emerging" rather than mainstream, and what kind of problems are they trying to solve that classical computers struggle with?